Systematic Experimentation using Masked Autoencoder (MAE) on CIFAR-10

We reproduce the Masked Autoencoder (MAE) framework (He et al., 2021) on CIFAR-10 (32×32), experimenting across three axes: masking ratio, patch size, and encoder capacity. Our implementation follows the original asymmetric encoder-decoder design — a Vision Transformer (ViT) encoder (Dosovitskiy et al., 2020) operating only on visible patches, with a lightweight decoder reconstructing masked patches using per-patch normalised MSE loss. Training uses AdamW (β_1=0.9, β_2=0.95) with cosine LR decay and warmup, directly mirroring He et al. (2021). The approach is conceptually rooted in BERT-style masked language modelling (Devlin et al., 2018), adapted from text to vision: randomly mask tokens and train the model to reconstruct them from context alone. The key differences from He et al. (2021) are scale — we use CIFAR-10 vs ImageNet-1K, 50 epochs vs 1600, and models up to 20M parameters vs 632M.

1.1 Experiment 1 – Masking Ratio

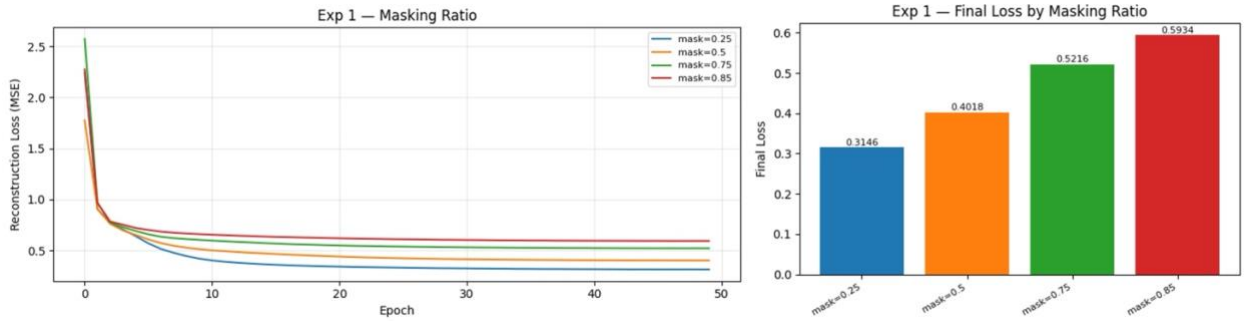

Figure 1 Reconstruction Loss in varying Masking Ratio

Masking ratios of 0.25, 0.50, 0.75, and 0.85 were tested with patch=4×4 and encoder=Small. Final losses were 0.3146, 0.4018, 0.5216, and 0.5934 respectively — monotonically increasing with masking ratio, exactly as He et al. (2021) predicts. Lower masking means more visible context, making reconstruction easier and MSE lower. However, mask=0.25 is not optimal for representation learning: the task is too easy, encouraging texture interpolation rather than semantic understanding. He et al. (2021) recommends 0.75 specifically because it forces the model to reason about global structure from sparse context — a harder task that produces richer learned representations.

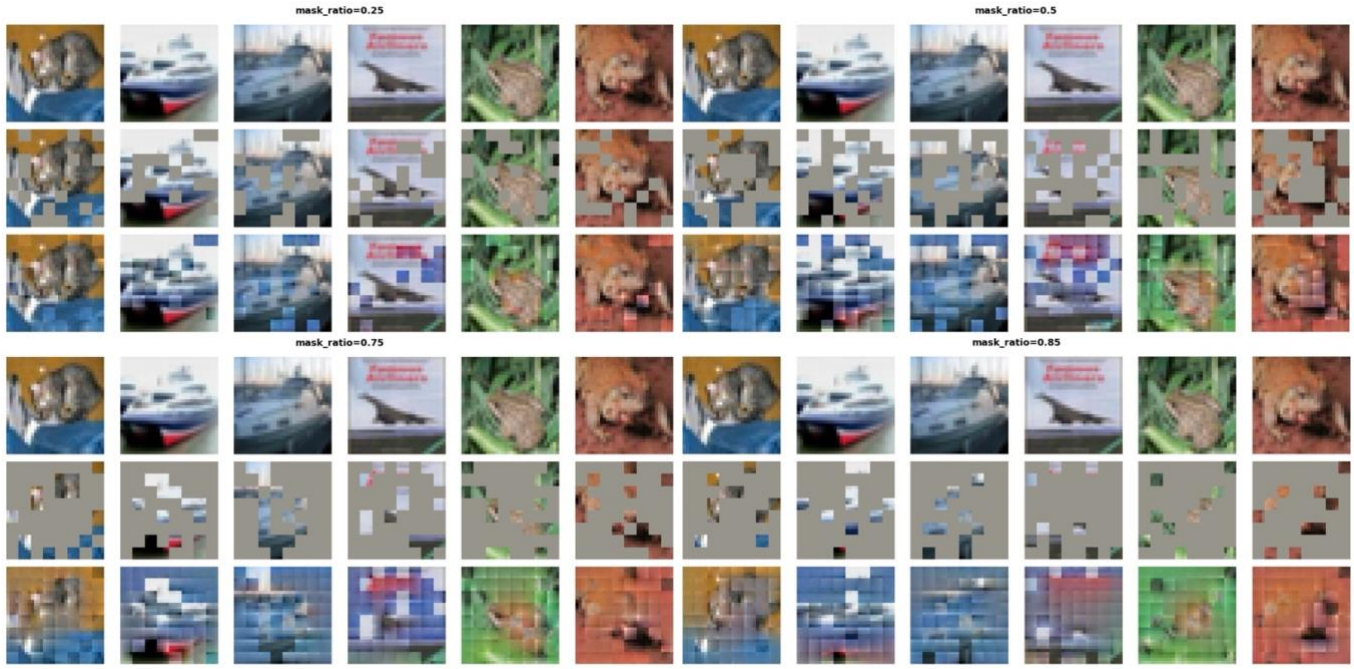

Figure 2 Reconstructed Images for different Masking Ratio

1.2 Experiment 2 – Patch Size

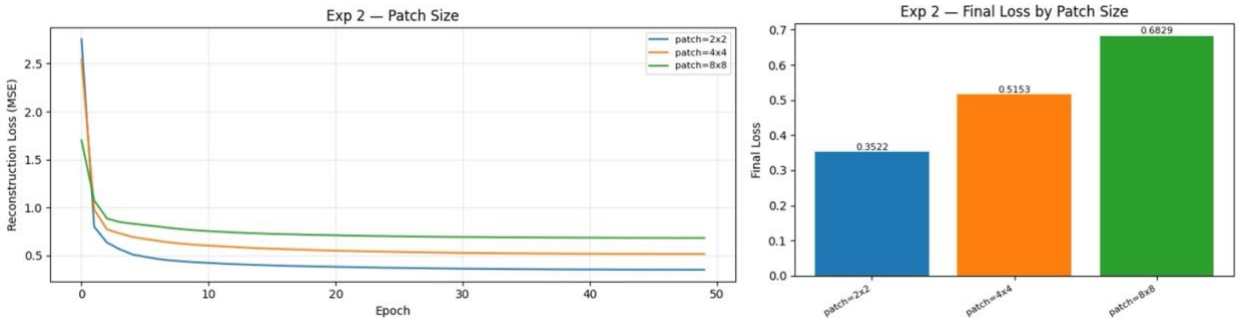

Figure 3 Reconstruction Loss in varying Patch Size

Patch sizes of 2×2 (256 tokens), 4×4 (64 tokens), and 8×8 (16 tokens) were tested at mask=0.75. Final losses were 0.3522, 0.5153, and 0.6829. Patch=2×2 achieves the lowest loss because fine-grained tokens are individually easy to reconstruct, and abundant visible context remains even at 75% masking. Patch=8×8 is the worst — only 4 patches remain visible at 75% masking on a 32×32 image, leaving the decoder with almost no spatial information. Despite its low loss, Patch=2×2 is not theory-optimal: Patch=4×4 (64 tokens) is proportionally equivalent to the 16×16 patches used by He et al. (2021) on 224×224 images, representing the best balance between task difficulty and spatial context for a 32×32 image. Consistent with Dosovitskiy et al. (2020), patch granularity fundamentally governs the richness of the token sequence fed to the transformer encoder.

Figure 4 Reconstructed Images for different Patch Sizes

1.3 Experiment 3 – Encoder Size

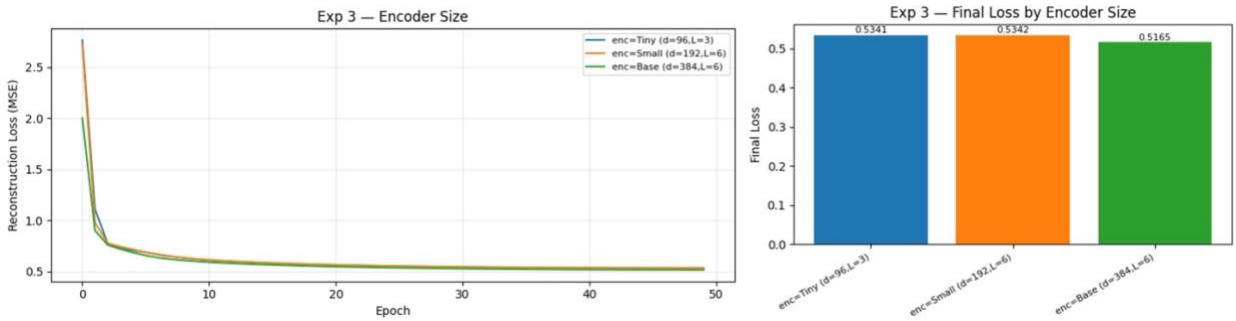

Figure 5 Reconstruction Loss by varying Encoder Size

Tiny (96d, 3L), Small (192d, 6L), and Base (384d, 6L) encoders were tested at mask=0.75, patch=4×4. Final losses were 0.5341, 0.5342, and 0.5165. Tiny and Small are virtually identical (Δ=0.0001), while Base achieves a modest 3.3% improvement. The near-zero gap between Tiny and Small reflects a data-limited regime: CIFAR-10's low resolution and 50 epochs of training do not provide sufficient learning signal to differentiate encoder capacities. Base's advantage confirms that capacity matters, but the scalability

claim of He et al. (2021) — that larger models benefit disproportionately from MAE pre-training — fully manifests only with longer training on harder, higher-resolution data.

Figure 6 Reconstructed Images by different Encoder Sizes

1.4 Conclusion

The loss-optimal configuration is mask=0.25, patch=2×2, encoder=Base — but this is not the recommended configuration. The theory-optimal setting is mask=0.75, patch=4×4, encoder=Base, directly mirroring the validated choices of He et al. (2021). The central takeaway is that reconstruction loss and representation quality are not the same thing (He et al., 2021): the harder the reconstruction task, the richer the learned visual representations, even at the cost of higher MSE. Future work should add linear probing evaluation to empirically confirm this on downstream CIFAR-10 classification.